Project1_RomainETIENNE_AnirudhSUDHIR

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1 Course assessment: project part "stock data" (M. Garcin)

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1.1 Numerical processing of financial data

Objective The goal of this project is to assess the risk and the performance of a simple portfolio.

- Please write commented code. You can also use text cells for your comments and explanations.
- The code must follow the good coding principles (it must be clean, readable, etc.).
- Graphs must have a title, a legend and their axes must be labeled.
- At the end, submit your notebooks **and a pdf file of 5 to 8 pages** into the Moodle Assignment you will find in the section "Assessment: projects". This separate pdf file is intended to be a (very) short paper : a standalone document that summarizes the work you have carried out in the notebooks and the results you have obtained (being they positive or negative), with figures. It has to be self-contained in the sense that one has to be able to read your report and understand its content without having to look into your notebooks, which will be checked afterwards.

[1]:
```
import numpy as np
import math
import matplotlib.pyplot as plt
import pandas as pd
import os
import sys
import time
```

[2]:
```
from rich.progress import Progress
import seaborn as sns
```

[3]:
```
# --- Figure saving setup ---
# Figures are exported to report/figures/ in both PDF (LaTeX) and PNG (preview).
# Run this cell before all others to ensure the directory exists.

FIGURES_DIR = os.path.join("report", "figures")
os.makedirs(FIGURES_DIR, exist_ok=True)
```

```
def save_fig(fig, name: str, dpi: int = 150) -> None:
    """Save a matplotlib figure as PDF (vector, for LaTeX) and PNG (raster, for␣
↪preview)."""
    fig.savefig(os.path.join(FIGURES_DIR, f"{name}.pdf"), bbox_inches="tight")
    fig.savefig(os.path.join(FIGURES_DIR, f"{name}.png"), bbox_inches="tight",␣
↪dpi=dpi)
```

2 Dataset

You will use the dataset (Excel files) of one-minute observations (OHLC prices) provided in Moodle for TP1 (Session 1). The dataset consists in the 40 constituents of the French stock index CAC 40 (as of 2022/2023). The biggest capitalizations are: LVMH (MC), L'Oréal (OR), TotalEnergies (TTE), Sanofi (SAN), Hermès (RMS), Airbus (AIR)…

You have first to restrict arbitrarily your investment universe to ten of these stocks and work with this universe for all the project.

The time range is September 2022-March 2023. Pay attention to the fact that some stocks don't have a price every minute (less liquid, like SW): avoid them.

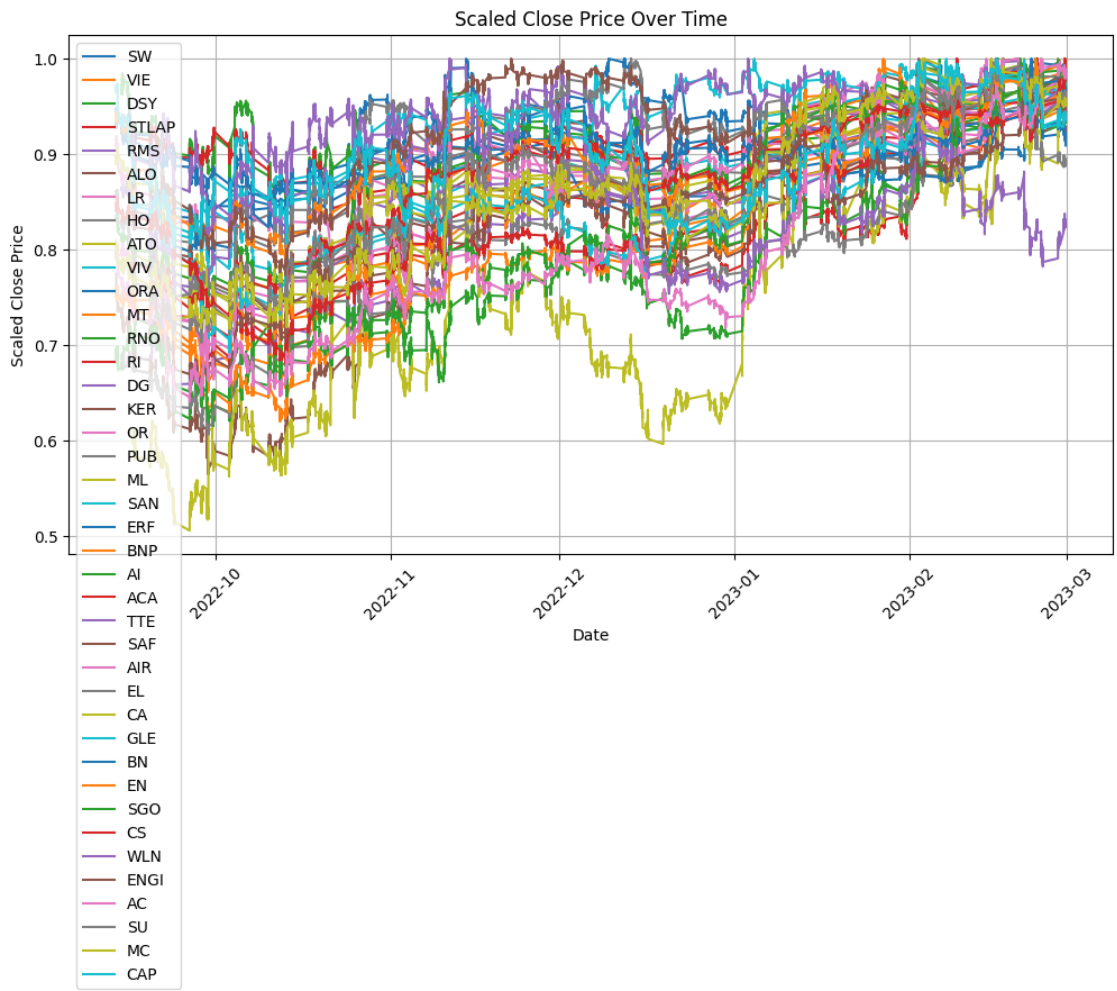
[4]:
```
# %pip install openpyxl  # if needed

begin_date = pd.to_datetime("2022-09-01")
end_date = pd.to_datetime("2023-03-01")

folder_path = 'data/'
df = pd.DataFrame()

with Progress() as p:
    t = p.add_task("[red]Processing files...", total=len(os.
↪listdir(folder_path)))

    for i, filename in enumerate(os.listdir(folder_path)):
        if filename.endswith('.xlsx'):
            file_path = os.path.join(folder_path, filename)

            sub_df = pd.read_excel(file_path, engine='openpyxl', header=1)

            # Enforcing convert to datetime
            if pd.api.types.is_numeric_dtype(sub_df['Dates']):
                sub_df['Dates'] = pd.to_datetime(sub_df['Dates'], unit='d',␣
↪origin='1899-12-30', errors='coerce')  # type: ignore

            sub_df = sub_df[(sub_df['Dates'] >= begin_date) & (sub_df['Dates']␣
↪<= end_date)]

            sub_df['Ticker'] = filename.split('_')[0]
```

```
            df = pd.concat([df, sub_df], ignore_index=True)
            p.update(t, advance=1)

print(df.head())
print(df.tail())
```

Output()

```
                    Dates   Open  Close   High    Low Ticker
0 2022-09-19 09:02:00.005  80.70  80.74  80.76  80.52     SW
1 2022-09-19 09:04:00.000  80.60  80.30  80.60  80.30     SW
2 2022-09-19 09:06:00.000  80.36  80.34  80.38  80.32     SW
3 2022-09-19 09:07:00.000  80.30  80.38  80.38  80.30     SW
4 2022-09-19 09:09:00.000  80.40  80.38  80.40  80.38     SW
                      Dates    Open   Close    High     Low Ticker
2256517 2023-02-28 17:27:00  178.70  178.75  178.80  178.70    CAP
2256518 2023-02-28 17:28:00  178.70  178.70  178.70  178.65    CAP
2256519 2023-02-28 17:29:00  178.70  178.75  178.75  178.70    CAP
2256520 2023-02-28 17:35:00  177.85  177.85  177.85  177.85    CAP
2256521 2023-02-28 17:37:00  177.85  177.85  177.85  177.85    CAP
```

[5]:
```
plt.figure(figsize=(12, 6))

for ticker in df["Ticker"].unique():
    grouped = df[df["Ticker"] == ticker]
    plt.plot(grouped["Dates"], grouped["Close"]/grouped["Close"].max(),␣
 ↪label=f"{ticker}")

plt.title(f"Scaled Close Price Over Time")
plt.xlabel("Date")
plt.xticks(rotation=45)
plt.ylabel("Scaled Close Price")
plt.legend()
plt.grid()
save_fig(plt.gcf(), "fig_00_scaled_close")
plt.show()
```



[6]:
```
# --- Ticker selection ---
# We restrict to 10 CAC 40 stocks that are:
#   (1) Liquid (all appear in the top half of the dataset by row count)
#   (2) Diversified: one primary sector each, so no single theme dominates
#
# Ticker | Company              | Sector
# -------|----------------------|----------------------------
# MC     | LVMH                 | Luxury goods
# TTE    | TotalEnergies        | Energy
# SAN    | Sanofi               | Healthcare / Pharma
# AIR    | Airbus               | Aerospace / Defence
# BNP    | BNP Paribas          | Banking
# CS     | AXA                  | Insurance
# SU     | Schneider Electric   | Industrial / Energy transition
# DSY    | Dassault Systèmes    | Enterprise Software
# MT     | ArcelorMittal        | Materials / Steel
```

```
# ORA     | Orange                | Telecom (defensive, dividend)

SELECTED_TICKERS = ["MC", "TTE", "SAN", "AIR", "BNP", "CS", "SU", "DSY", "MT",␣
 ↪"ORA"]

df_selected = df[df["Ticker"].isin(SELECTED_TICKERS)].copy()

print(f"Selected tickers : {SELECTED_TICKERS}")
print(f"Date range       : {df_selected['Dates'].min()} → {df_selected['Dates'].
 ↪max()}")
print("\nRow count per ticker (proxy for liquidity):")
print(df_selected.groupby("Ticker").size().sort_values(ascending=False).
 ↪to_string())
```

```
Selected tickers : ['MC', 'TTE', 'SAN', 'AIR', 'BNP', 'CS', 'SU', 'DSY', 'MT',
'ORA']
Date range       : 2022-09-13 09:00:00 → 2023-02-28 17:38:00

Row count per ticker (proxy for liquidity):
Ticker
MC     61371
BNP    60911
AIR    60769
CS     60150
ORA    59821
TTE    59410
SU     59394
DSY    59276
SAN    58861
MT     58454
```

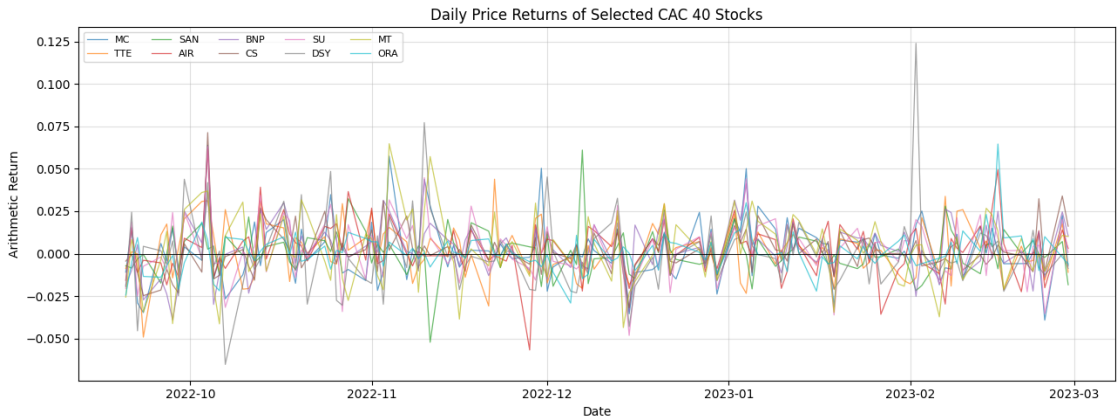
3 Creating a porfolio (Part 1)

- **1.a/** Split your dataset of prices in separate business days, from day 1 to day n. Calculate and put in a dataframe the price returns of the 10 stocks at a daily time scale.

[7]:
```
# 1.a - Extract the last Close price of each (day, ticker) pair, then compute
#       arithmetic daily returns: R_{i,j} = (P_{i,j} - P_{i-1,j}) / P_{i-1,j}

daily_close = (
    df_selected.copy()
    .assign(Date=df_selected["Dates"].dt.normalize())   # strip intraday time
    .groupby(["Date", "Ticker"])["Close"]
    .last()                                              # last quote of the day
    .unstack("Ticker")[SELECTED_TICKERS]                 # consistent column␣
 ↪order
)
```

```python
daily_returns = daily_close.pct_change().dropna()
daily_returns.index.name = "Date"

n = len(daily_close)        # number of price days
print(f"Price days (n)             : {n}")
print(f"Return days (n-1)          : {len(daily_returns)}")
print(f"\nFirst 3 rows of daily_returns:")
print(daily_returns.round(4).head(3))

# Visualise
fig, ax = plt.subplots(figsize=(13, 5))
for ticker in daily_returns.columns:
    ax.plot(daily_returns.index, daily_returns[ticker], alpha=0.7, linewidth=0.
 ↪9, label=ticker)
ax.axhline(0, color="black", linewidth=0.6)
ax.set_title("Daily Price Returns of Selected CAC 40 Stocks")
ax.set_xlabel("Date")
ax.set_ylabel("Arithmetic Return")
ax.legend(ncol=5, fontsize=8)
ax.grid(True, alpha=0.4)
plt.tight_layout()
save_fig(fig, "fig_01_daily_returns")
plt.show()
```

```
Price days (n)             : 120
Return days (n-1)          : 115

First 3 rows of daily_returns:
Ticker          MC     TTE     SAN     AIR     BNP      CS      SU     DSY  \
Date
2022-09-20 -0.0109 -0.0087 -0.0071 -0.0157 -0.0185 -0.0044 -0.0206 -0.0194
2022-09-21  0.0151  0.0011 -0.0085  0.0152 -0.0098 -0.0006  0.0186  0.0244
2022-09-22 -0.0291 -0.0002 -0.0030 -0.0110  0.0053 -0.0114 -0.0250 -0.0455

Ticker          MT     ORA
Date
2022-09-20 -0.0256 -0.0244
2022-09-21  0.0083 -0.0051
2022-09-22 -0.0039  0.0094
```

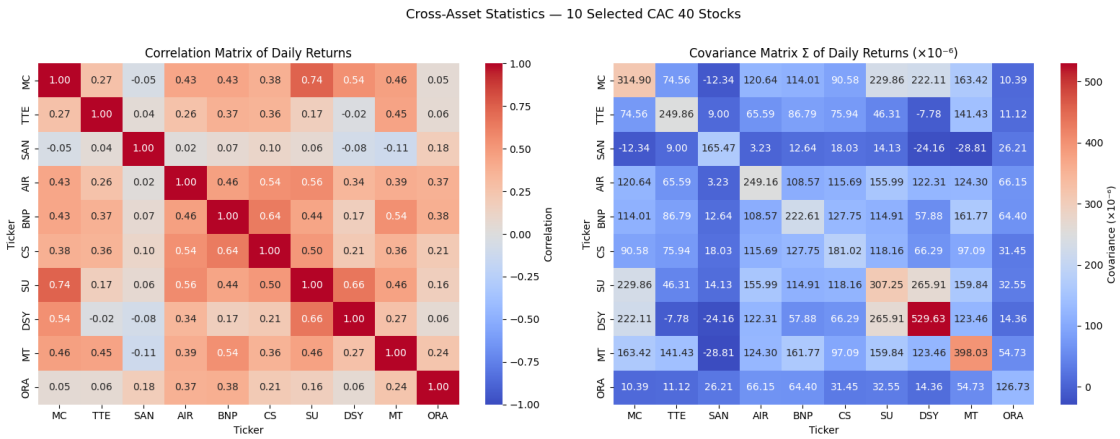


- **1.b/** Calculate both the correlation matrix and the covariance matrix Σ of the daily price returns of the 10 stocks.

[8]:
```
# 1.b - Correlation matrix and covariance matrix Σ of daily price returns.
#       Σ is kept at the daily scale (not annualised) to match the daily
#       return projections used in the Markowitz formula.

corr_matrix = daily_returns.corr()
cov_matrix  = daily_returns.cov()   # Σ (daily)

print("Correlation matrix:")
print(corr_matrix.round(3).to_string())
print("\nCovariance matrix Σ (daily, ×10 ):")
print((cov_matrix * 1e6).round(2).to_string())

fig, axes = plt.subplots(1, 2, figsize=(16, 6))

sns.heatmap(corr_matrix, annot=True, fmt=".2f", cmap="coolwarm",
            vmin=-1, vmax=1, ax=axes[0], cbar_kws={"label": "Correlation"})
axes[0].set_title("Correlation Matrix of Daily Returns")

sns.heatmap(cov_matrix * 1e6, annot=True, fmt=".2f", cmap="coolwarm",
            ax=axes[1], cbar_kws={"label": "Covariance (×10 )"})
axes[1].set_title("Covariance Matrix Σ of Daily Returns (×10 )")

plt.suptitle("Cross-Asset Statistics - 10 Selected CAC 40 Stocks", fontsize=13,␣
 ↪y=1.01)
plt.tight_layout()
save_fig(fig, "fig_02_corr_cov_matrices")
plt.show()
```

```
Correlation matrix:
Ticker     MC    TTE    SAN    AIR    BNP     CS     SU    DSY     MT    ORA
```

```
Ticker
MC      1.000  0.266 -0.054  0.431  0.431  0.379  0.739  0.544  0.462  0.052
TTE     0.266  1.000  0.044  0.263  0.368  0.357  0.167 -0.021  0.448  0.062
SAN    -0.054  0.044  1.000  0.016  0.066  0.104  0.063 -0.082 -0.112  0.181
AIR     0.431  0.263  0.016  1.000  0.461  0.545  0.564  0.337  0.395  0.372
BNP     0.431  0.368  0.066  0.461  1.000  0.636  0.439  0.169  0.543  0.383
CS      0.379  0.357  0.104  0.545  0.636  1.000  0.501  0.214  0.362  0.208
SU      0.739  0.167  0.063  0.564  0.439  0.501  1.000  0.659  0.457  0.165
DSY     0.544 -0.021 -0.082  0.337  0.169  0.214  0.659  1.000  0.269  0.055
MT      0.462  0.448 -0.112  0.395  0.543  0.362  0.457  0.269  1.000  0.244
ORA     0.052  0.062  0.181  0.372  0.383  0.208  0.165  0.055  0.244  1.000

Covariance matrix Σ (daily, ×10 ):
Ticker      MC     TTE     SAN     AIR     BNP      CS      SU     DSY      MT
ORA
Ticker
MC      314.90   74.56  -12.34  120.64  114.01   90.58  229.86  222.11  163.42
10.39
TTE      74.56  249.86    9.00   65.59   86.79   75.94   46.31   -7.78  141.43
11.12
SAN     -12.34    9.00  165.47    3.23   12.64   18.03   14.13  -24.16  -28.81
26.21
AIR     120.64   65.59    3.23  249.16  108.57  115.69  155.99  122.31  124.30
66.15
BNP     114.01   86.79   12.64  108.57  222.61  127.75  114.91   57.88  161.77
64.40
CS       90.58   75.94   18.03  115.69  127.75  181.02  118.16   66.29   97.09
31.45
SU      229.86   46.31   14.13  155.99  114.91  118.16  307.25  265.91  159.84
32.55
DSY     222.11   -7.78  -24.16  122.31   57.88   66.29  265.91  529.63  123.46
14.36
MT      163.42  141.43  -28.81  124.30  161.77   97.09  159.84  123.46  398.03
54.73
ORA      10.39   11.12   26.21   66.15   64.40   31.45   32.55   14.36   54.73
126.73
```

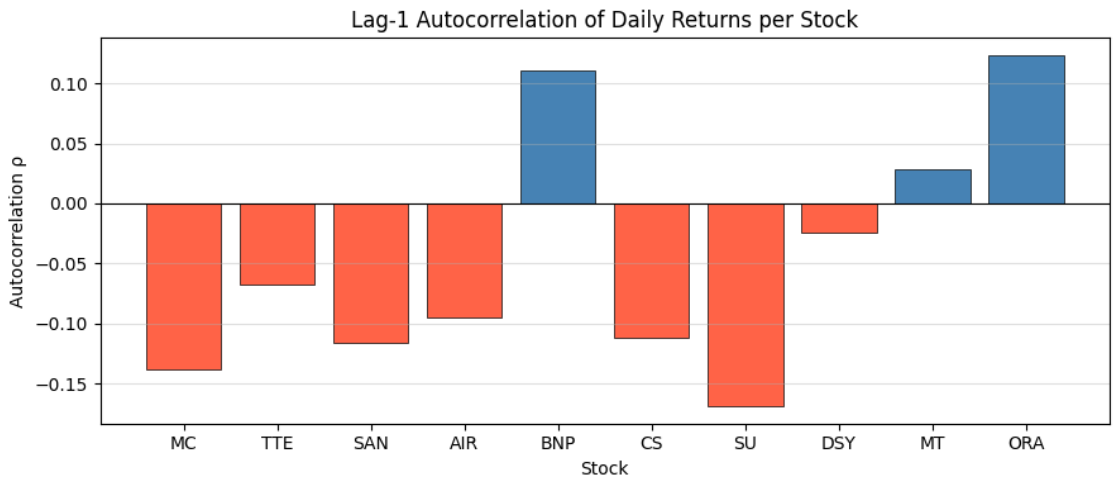


- **1.c/** For each stock, calculate the autocorrelation of the daily time series of returns, for a lag of one day.

[9]:
```
# 1.c - Lag-1 autocorrelation _j of each stock's daily return series.

autocorr = daily_returns.apply(lambda s: s.autocorr(lag=1))

print("Lag-1 autocorrelation _j per stock:")
print(autocorr.round(4).to_string())

fig, ax = plt.subplots(figsize=(9, 4))
colors = ["steelblue" if v >= 0 else "tomato" for v in autocorr.values]
ax.bar(autocorr.index, autocorr.values, color=colors, edgecolor="black",␣
↪linewidth=0.5)
ax.axhline(0, color="black", linewidth=0.8)
ax.set_title("Lag-1 Autocorrelation of Daily Returns per Stock")
ax.set_xlabel("Stock")
ax.set_ylabel("Autocorrelation ")
ax.grid(True, axis="y", alpha=0.4)
plt.tight_layout()
save_fig(fig, "fig_03_autocorrelation")
plt.show()
```

```
Lag-1 autocorrelation _j per stock:
Ticker
MC    -0.1381
TTE   -0.0671
SAN   -0.1164
AIR   -0.0946
BNP    0.1103
CS    -0.1113
SU    -0.1684
DSY   -0.0244
```

```
MT      0.0280
ORA     0.1230
```

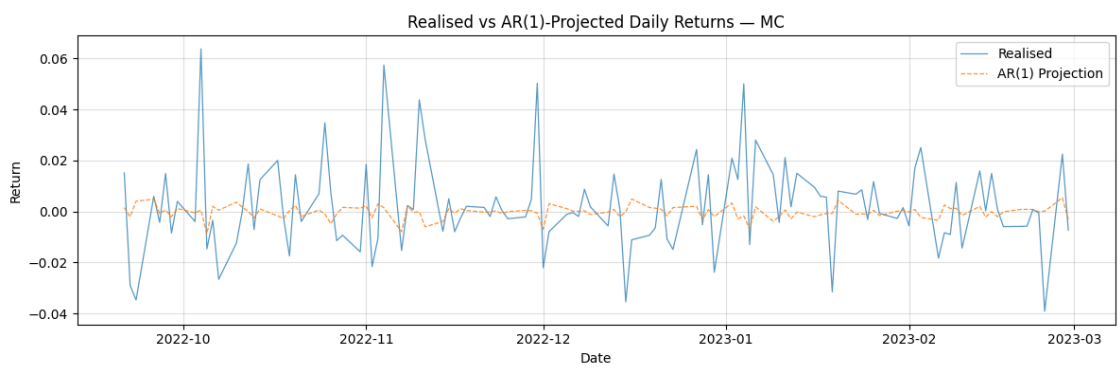


- **1.d/** Using only the autocorrelations calculated above and the observation of the price returns at day i (for $1 \leq i \leq n-1$), propose a projection $R_{i+1,j}$ of the future price return at day $i+1$ for each stock j ($1 \leq j \leq 10$). Justify briefly the relevance of the method.

[10]:
```
# 1.d - AR(1) projection: R_{i+1, j} = _j · R_{i, j}
#
# Justification: under a zero-mean AR(1) model,  R_{i+1} =   · R_i + _{i+1},
# the best linear one-step-ahead forecast given R_i is  E[R_{i+1} | R_i] =   ·␣
↪R_i.
# The autocorrelation _j is the only linear link between consecutive returns,
# so even though it may be small, it represents the maximum extractable
# linear signal from a single lag.

rho = autocorr.values   # shape (10,)

# For day index k in daily_returns (0-based):
#   observed_returns.iloc[k] corresponds to day k
#   the projection targets day k+1 (i.e. daily_returns.index[k+1])
observed_returns  = daily_returns.iloc[:-1]             # days 0 … n-3
projected_values  = observed_returns.values * rho       # broadcast  along␣
↪columns

projected_returns_df = pd.DataFrame(
    projected_values,
    index=daily_returns.index[1:],   # prediction target dates: day 1 … day n-2
    columns=SELECTED_TICKERS,
)
```

```python
print("Projected returns (first 3 rows):")
print(projected_returns_df.round(5).head(3))

# Visual check: realised vs AR(1) projection for one example stock
ticker_example = "MC"
fig, ax = plt.subplots(figsize=(12, 4))
ax.plot(daily_returns.index[1:], daily_returns.loc[daily_returns.index[1:],␣
 ↪ticker_example],
        label="Realised", alpha=0.75, linewidth=0.9)
ax.plot(projected_returns_df.index, projected_returns_df[ticker_example],
        label="AR(1) Projection", alpha=0.85, linewidth=0.9, linestyle="--")
ax.set_title(f"Realised vs AR(1)-Projected Daily Returns - {ticker_example}")
ax.set_xlabel("Date")
ax.set_ylabel("Return")
ax.legend()
ax.grid(True, alpha=0.4)
plt.tight_layout()
save_fig(fig, "fig_04_ar1_projection")
plt.show()
```

```
Projected returns (first 3 rows):
                 MC      TTE      SAN      AIR      BNP       CS       SU  \
Date
2022-09-21  0.00151  0.00059  0.00083  0.00149 -0.00205  0.00049  0.00347
2022-09-22 -0.00209 -0.00008  0.00099 -0.00144 -0.00108  0.00007 -0.00313
2022-09-23  0.00401  0.00001  0.00035  0.00104  0.00059  0.00127  0.00422

                DSY       MT      ORA
Date
2022-09-21  0.00047 -0.00072 -0.00300
2022-09-22 -0.00060  0.00023 -0.00063
2022-09-23  0.00111 -0.00011  0.00116
```



- **1.e/** Using the above projection as a proxy for the expected return, calculate the Markowitz

portfolio such that the sum of weights is equal to 1, that is the optimal vector of weights for day $i+1$ is

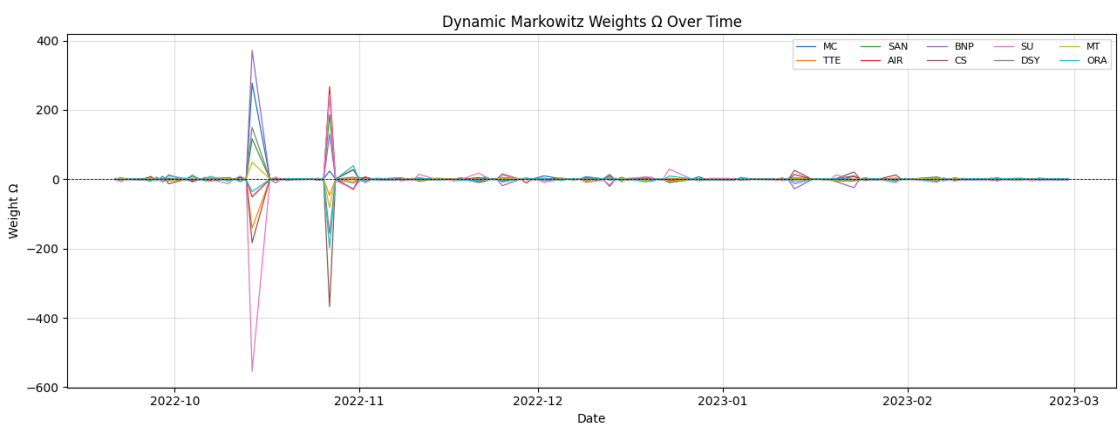
$$\Omega_{i+1} = \frac{1}{\mathbf{1}_{10}'\Sigma^{-1}R_{i+1}}\Sigma^{-1}R_{i+1},$$

where $R_{i+1} = (R_{i+1,1}, ..., R_{i+1,10})'$ and $\mathbf{1}_{10}$ is a vector of ones of size 10.

[11]:
```
# 1.e - Dynamic Markowitz weights (sum-to-one constraint)
#
# For each day d in projected_returns_df.index:
#   Ω(d) = Σ⁻¹ R(d) / (1' Σ⁻¹ R(d))
#
# Σ is estimated once on the full sample (see 1.g for the implied bias).

Sigma     = cov_matrix.values        # (10×10), daily covariance
Sigma_inv = np.linalg.inv(Sigma)     # (10×10)


def markowitz_weights(R_proj: np.ndarray, Sigma_inv: np.ndarray) -> np.ndarray:
    """Compute sum-to-1 Markowitz weights given projected returns and Σ⁻¹."""
    raw   = Sigma_inv @ R_proj
    denom = raw.sum()
    if np.abs(denom) < 1e-12:
        # Degenerate case: projected returns are near-zero → equal weights␣
 ↪fallback
        return np.full(len(R_proj), 1.0 / len(R_proj))
    return raw / denom


dynamic_weights = np.array([
    markowitz_weights(projected_returns_df.iloc[i].values, Sigma_inv)  # type:␣
 ↪ignore
    for i in range(len(projected_returns_df))
])

dynamic_weights_df = pd.DataFrame(
    dynamic_weights,
    index=projected_returns_df.index,
    columns=SELECTED_TICKERS,
)

# Sanity check: all rows must sum to 1
assert np.allclose(dynamic_weights_df.sum(axis=1), 1.0), "Weight sums deviate␣
 ↪from 1!"
print("Weight-sum check passed (all rows sum to 1.0)")
print("\nDynamic weights - first 3 days:")
print(dynamic_weights_df.round(3).head(3))
```

```python
# Time series of Markowitz weights
fig, ax = plt.subplots(figsize=(13, 5))
for ticker in dynamic_weights_df.columns:
    ax.plot(dynamic_weights_df.index, dynamic_weights_df[ticker],
            linewidth=0.9, label=ticker)
ax.axhline(0, color="black", linewidth=0.6, linestyle="--")
ax.set_title("Dynamic Markowitz Weights Ω Over Time")
ax.set_xlabel("Date")
ax.set_ylabel("Weight Ω")
ax.legend(ncol=5, fontsize=8)
ax.grid(True, alpha=0.4)
plt.tight_layout()
save_fig(fig, "fig_05_dynamic_weights")
plt.show()
```

```
Weight-sum check passed (all rows sum to 1.0)

Dynamic weights - first 3 days:
               MC    TTE    SAN    AIR    BNP     CS     SU    DSY     MT  \
Date
2022-09-21  0.408 -0.237 -0.426 -0.765  1.090  0.118 -2.035  0.727  0.098
2022-09-22  0.204 -0.865  3.143 -0.359 -2.256  4.333 -7.155  2.476  2.885
2022-09-23  0.899  0.007 -0.269 -0.602 -0.635  0.270  1.519 -0.734 -0.564

              ORA
Date
2022-09-21  2.021
2022-09-22 -1.406
2022-09-23  1.108
```

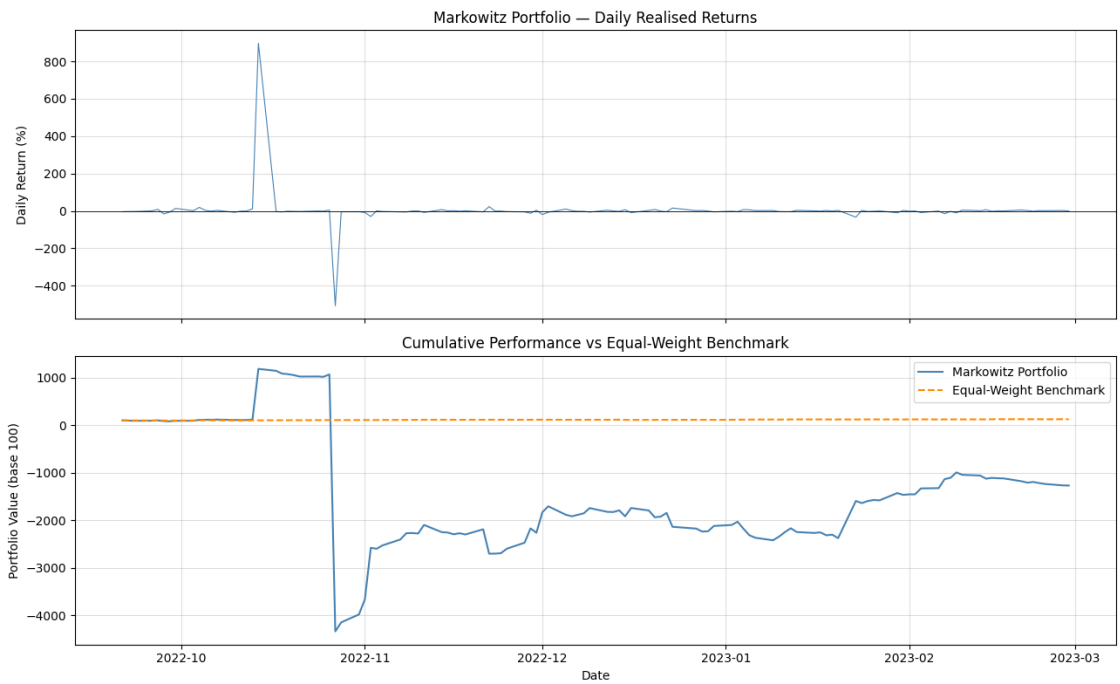


- **1.f/** Add in your dataframe the time series of realized price returns for your portfolio, between day 2 and day n, using the dynamic weights calculated above.

[12]:
```
# 1.f - Realised portfolio return on each day d:
#   r_ptf(d) = Ω(d)' · R_realised(d)
# where R_realised(d) are the actual observed returns on day d.

realised_aligned = daily_returns.loc[dynamic_weights_df.index]   # align index

portfolio_returns = pd.Series(
    (dynamic_weights_df.values * realised_aligned.values).sum(axis=1),
    index=dynamic_weights_df.index,
    name="Portfolio",
)

# Append the portfolio column to daily_returns for later use
daily_returns["Portfolio"] = portfolio_returns

print("Portfolio return summary:")
print(portfolio_returns.describe().round(5))

# --- Cumulative performance plot ---
initial_value = 100
ptf_value = initial_value * (1 + portfolio_returns).cumprod()

# Equal-weight index as benchmark
ew_returns = realised_aligned.mean(axis=1)
ew_value   = initial_value * (1 + ew_returns).cumprod()

fig, axes = plt.subplots(2, 1, figsize=(13, 8), sharex=True)

axes[0].plot(portfolio_returns.index, portfolio_returns * 100,
             linewidth=0.8, color="steelblue")
axes[0].axhline(0, color="black", linewidth=0.6)
axes[0].set_ylabel("Daily Return (%)")
axes[0].set_title("Markowitz Portfolio - Daily Realised Returns")
axes[0].grid(True, alpha=0.4)

axes[1].plot(ptf_value.index, ptf_value,
             linewidth=1.5, color="steelblue", label="Markowitz Portfolio")
axes[1].plot(ew_value.index, ew_value,
             linewidth=1.5, color="darkorange", linestyle="--",␣
 ↪label="Equal-Weight Benchmark")
axes[1].set_ylabel("Portfolio Value (base 100)")
axes[1].set_xlabel("Date")
axes[1].set_title("Cumulative Performance vs Equal-Weight Benchmark")
axes[1].legend()
axes[1].grid(True, alpha=0.4)

plt.tight_layout()
```

```
save_fig(fig, "fig_06_portfolio_performance")
plt.show()
```

```
Portfolio return summary:
count    114.00000
mean       0.02690
std        0.97169
min       -5.07370
25%       -0.03590
50%       -0.00376
75%        0.02462
max        8.96755
Name: Portfolio, dtype: float64
```



- **1.g/** What is the main methodological bias we had in the construction of this portfolio that would affect the quality of a backtest?

Answer — 1.g

The main methodological bias is **look-ahead bias** (also called *full-sample bias* or *data snooping*).

The covariance matrix Σ is estimated on the **entire** price series from September 2022 to March 2023, including observations that lie strictly in the future relative to any trading day i. Because Σ implicitly encodes future co-movement structure, the Markowitz weights Ω_{i+1} are computed with information that would never be available in a real out-of-sample deployment, artificially inflating apparent portfolio quality.

A rigorous backtest would estimate Σ from a **rolling or expanding window** of strictly past

data (e.g. all days up to and including day i), ensuring no future information leaks into the weight computation.

4 Risk and performance evaluation (Part 2)

- **2.a/** Determine the ex-ante (that is using $R_{i+1,j}$ as the return) and ex-post (that is the real return observed after each trading day) Sharpe ratio of your portfolio (consider 0 as the risk-free rate).

[13]:
```
# 2.a - Sharpe ratio (risk-free rate = 0)
#
# Ex-ante : based on the projected (expected) portfolio return each day,
#             i.e. r̂_ptf(d) = Ω(d)' · R(d)
# Ex-post  : based on the realised portfolio return each day,
#             i.e. r_ptf(d) = Ω(d)' · R_realised(d)

# Ex-ante projected portfolio return
ex_ante_ptf_returns = pd.Series(
    (dynamic_weights_df.values * projected_returns_df.values).sum(axis=1),
    index=dynamic_weights_df.index,
    name="Ex-Ante Portfolio Return",
)


def sharpe_ratio(returns: pd.Series, risk_free: float = 0.0, annualise: bool =␣
 ↪False) -> float:
    """Sharpe ratio = mean(excess) / std(excess).  Annualise with √252 if␣
 ↪requested."""
    excess = returns - risk_free
    sr = excess.mean() / excess.std() if excess.std() > 0 else np.nan
    return sr * np.sqrt(252) if annualise else sr


sr_ante_daily = sharpe_ratio(ex_ante_ptf_returns)
sr_post_daily = sharpe_ratio(portfolio_returns)
sr_ante_ann   = sharpe_ratio(ex_ante_ptf_returns, annualise=True)
sr_post_ann   = sharpe_ratio(portfolio_returns,   annualise=True)

print(f"{'':30s} {'Daily':>10}  {'Annualised':>12}")
print(f"{'Ex-ante Sharpe ratio':30s} {sr_ante_daily:>10.4f}  {sr_ante_ann:>12.
 ↪4f}")
print(f"{'Ex-post Sharpe ratio':30s} {sr_post_daily:>10.4f}  {sr_post_ann:>12.
 ↪4f}")
```

```
                                Daily    Annualised
Ex-ante Sharpe ratio           0.0604        0.9595
Ex-post Sharpe ratio           0.0277        0.4394
```

- **2.b/** Calculate the ex-post upside potential ratio of your portfolio.

[14]:
```
# 2.b - Ex-post Upside Potential Ratio (UPR)
#
# UPR = E[max(R, 0)] / sqrt(E[min(R, 0)²])
#     = mean of positive returns / RMS of negative returns
#
# A ratio > 1 means the average gain is larger than the average loss magnitude;
# the higher the ratio, the more asymmetrically favourable the return profile.

r = portfolio_returns.values

positive_returns = r[r > 0]  # type: ignore
negative_returns = r[r < 0]  # type: ignore

upside = positive_returns.mean()  # type: ignore      # mean gain
downside = np.sqrt(
    (negative_returns**2).mean()  # type: ignore
)  # lower partial moment (RMS)

upr = upside / downside

print(f"Mean positive return  (upside)  : {upside:.6f}")
print(f"RMS of negative returns (downside): {downside:.6f}")
print(f"Upside Potential Ratio            : {upr:.4f}")
```

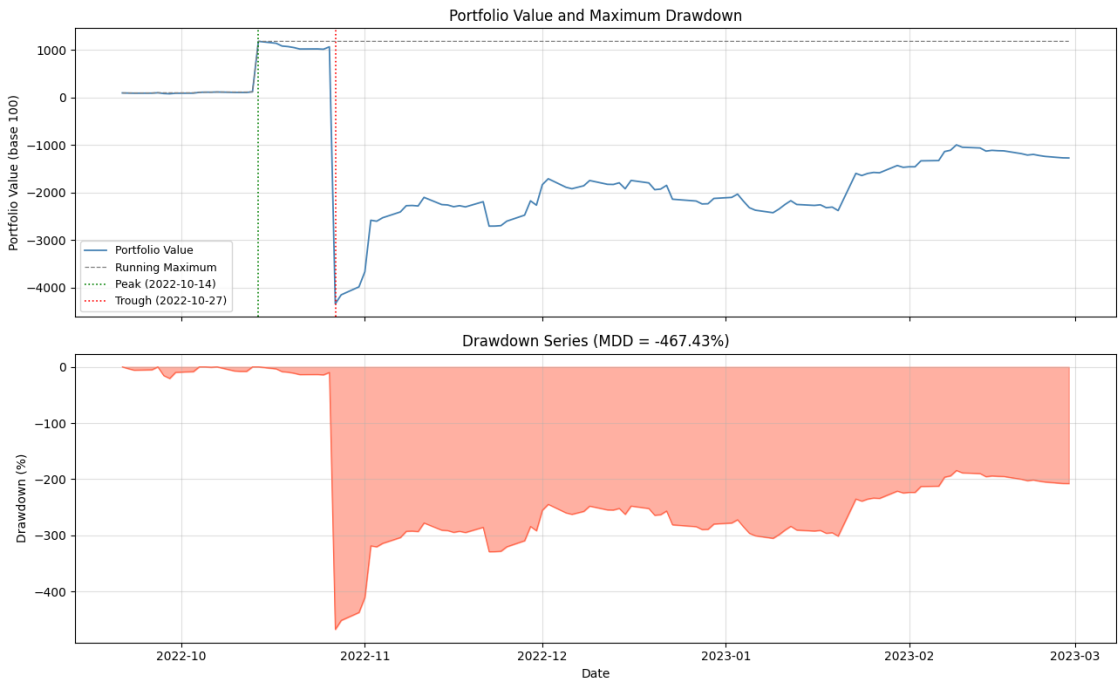
```
Mean positive return  (upside)  : 0.211555
RMS of negative returns (downside): 0.654557
Upside Potential Ratio            : 0.3232
```

- **2.c/** Calculate the maximum drawdown of your portfolio.

[15]:
```
# 2.c - Maximum Drawdown
#
# MDD = max_{t1 < t2}  (V(t1) - V(t2)) / V(t1)
#     = worst peak-to-trough percentage decline of the portfolio value.

def max_drawdown(values: pd.Series):
    """Return (mdd, peak_date, trough_date) for a series of portfolio values."""
    running_max  = values.cummax()
    drawdown     = (values - running_max) / running_max
    trough_date  = drawdown.idxmin()
    peak_date    = running_max.loc[:trough_date].idxmax()
    return drawdown.min(), peak_date, trough_date


mdd, peak_date, trough_date = max_drawdown(ptf_value)
```

```python
print(f"Maximum Drawdown : {100 * mdd:.2f}%")
print(f"Peak date        : {peak_date.date()}")  # type: ignore
print(f"Trough date      : {trough_date.date()}")  # type: ignore

# Visualise
running_max    = ptf_value.cummax()
drawdown_series = (ptf_value - running_max) / running_max

fig, axes = plt.subplots(2, 1, figsize=(13, 8), sharex=True)

axes[0].plot(ptf_value.index, ptf_value, color="steelblue", linewidth=1.3,␣
 ↪label="Portfolio Value")
axes[0].plot(running_max.index, running_max, color="grey", linewidth=0.9,␣
 ↪linestyle="--", label="Running Maximum")
axes[0].axvline(peak_date, color="green", linewidth=1.2, linestyle=":",␣
 ↪label=f"Peak ({peak_date.date()})")  # type: ignore
axes[0].axvline(trough_date, color="red", linewidth=1.2, linestyle=":",␣
 ↪label=f"Trough ({trough_date.date()})")  # type: ignore
axes[0].set_ylabel("Portfolio Value (base 100)")
axes[0].set_title("Portfolio Value and Maximum Drawdown")
axes[0].legend(fontsize=9)
axes[0].grid(True, alpha=0.4)

axes[1].fill_between(drawdown_series.index, drawdown_series * 100, 0,
                     color="tomato", alpha=0.5)
axes[1].plot(drawdown_series.index, drawdown_series * 100,
             color="tomato", linewidth=0.8)
axes[1].set_ylabel("Drawdown (%)")
axes[1].set_xlabel("Date")
axes[1].set_title(f"Drawdown Series (MDD = {100 * mdd:.2f}%)")
axes[1].grid(True, alpha=0.4)

plt.tight_layout()
save_fig(fig, "fig_07_drawdown")
plt.show()
```

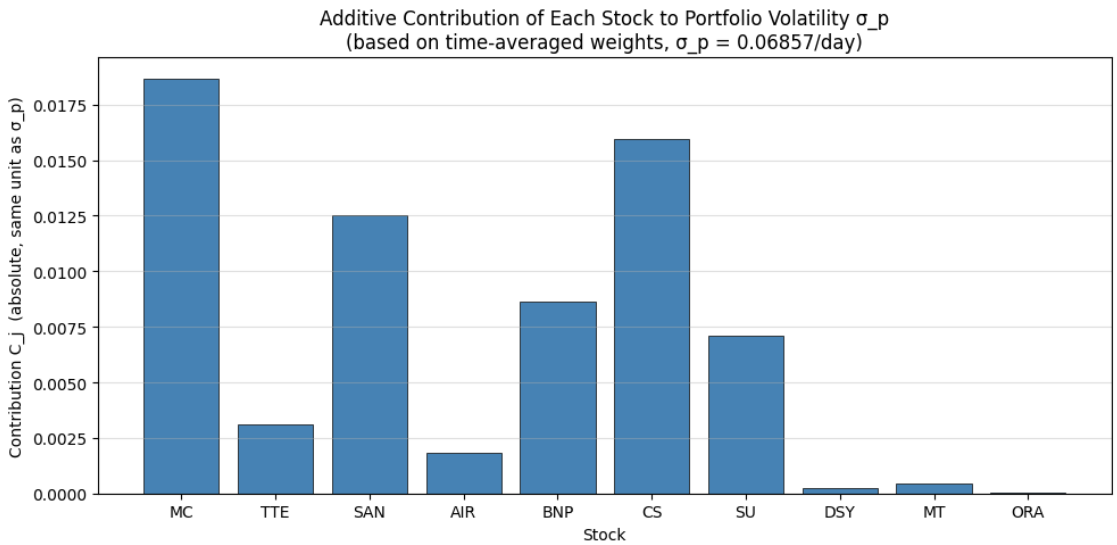
```
Maximum Drawdown : -467.43%
Peak date        : 2022-10-14
Trough date      : 2022-10-27
```



- **2.d/** Using the formula presented in the course, calculate the additive contribution of each of the 10 stocks to the volatility of the portfolio.

[16]:
```
# 2.d - Additive Contribution of Each Stock to Portfolio Volatility
#
# Portfolio variance decomposes exactly as:
#    ²_p = Ω' Σ Ω = Σ_j  Ω_j · (Σ Ω)_j
#
# Dividing by _p gives the additive decomposition of volatility:
#    _p = Σ_j  C_j,    with  C_j = Ω_j · (Σ Ω)_j / _p
#
# C_j > 0: stock j amplifies portfolio volatility.
# C_j < 0: stock j acts as a hedge (its weight is negative and its contribution␣
↪reduces _p).
# We use the time-averaged weight vector as the representative allocation.

mean_weights = dynamic_weights_df.mean().values        # shape (10,)
sigma_p_sq   = mean_weights @ Sigma @ mean_weights     # portfolio variance
sigma_p      = np.sqrt(sigma_p_sq)                     # portfolio daily␣
↪volatility

marginal  = Sigma @ mean_weights                       # (Σ Ω), shape (10,)
absolute_contributions = mean_weights * marginal / sigma_p     # C_j
pct_contributions      = 100 * absolute_contributions / sigma_p  # % of _p
```

```
vol_contrib_df = pd.DataFrame({
    "Mean Weight Ω_j":           mean_weights,
    "Marginal contrib (ΣΩ)_j":   marginal,
    "Additive contrib C_j":      absolute_contributions,
    "Share of  _p (%)":          pct_contributions,
}, index=SELECTED_TICKERS)

print(f"Portfolio daily volatility  _p  = {sigma_p:.6f}  "
      f"(annualised: {sigma_p * np.sqrt(252):.4f})")
print(f"Cross-check: Σ C_j             = {absolute_contributions.sum():.6f}    ␣
 ↪ _p")
print()
print(vol_contrib_df.round(6).to_string())

# Bar chart
fig, ax = plt.subplots(figsize=(10, 5))
colors = ["steelblue" if v >= 0 else "tomato" for v in absolute_contributions]
ax.bar(SELECTED_TICKERS, absolute_contributions,
       color=colors, edgecolor="black", linewidth=0.5)
ax.axhline(0, color="black", linewidth=0.6)
ax.set_title("Additive Contribution of Each Stock to Portfolio Volatility  _p\n"
             f"(based on time-averaged weights,  _p = {sigma_p:.5f}/day)")
ax.set_xlabel("Stock")
ax.set_ylabel("Contribution C_j  (absolute, same unit as  _p)")
ax.grid(True, axis="y", alpha=0.4)
plt.tight_layout()
save_fig(fig, "fig_08_vol_contributions")
plt.show()
```

```
Portfolio daily volatility  _p  = 0.068574  (annualised: 1.0886)
Cross-check: Σ C_j             = 0.068574     _p

     Mean Weight Ω_j  Marginal contrib (ΣΩ)_j  Additive contrib C_j  Share of
 _p (%)
MC          3.537580                 0.000362              0.018682
27.243361
TTE        -1.338332                -0.000160              0.003131
4.565606
SAN         2.772823                 0.000310              0.012547
18.296723
AIR         1.972992                 0.000063              0.001806
2.633876
BNP         3.054635                 0.000194              0.008623
12.574224
CS         -3.854890                -0.000284              0.015960
23.274311
SU         -3.261198                -0.000149              0.007094
10.345288
```

```
DSY        -0.142848                -0.000116              0.000241
0.351312
MT         -0.362463                -0.000085              0.000447
0.651785
ORA        -1.378298                -0.000002              0.000044
0.063514
```